Study Bot

J. Anithvithushan

1 . Project overview and explanation of memory implementation

- ⇨ It's a Study Bot, It's an AI-powered chatbot that helps users ask questions related to study topics and remembers previous conversations using a database. The bot could be able to store chat history, provide contextual responses.
- ⇨ This chatbot uses database memory, it store every message in MongoDB and rebuilds the conversation context each time the user sends a new question
- ⇨ Every chat turn is saved as a separate MongoDB document in the users collection.
- ⇨ For every user question, I insert two documents:
  - The user message
  - The assistant response
- ⇨ Document has
  - user_id - who this chat belongs to
  - role - "user" or "assistant"
  - message - actual text content

So the DB becomes their long-term memory log

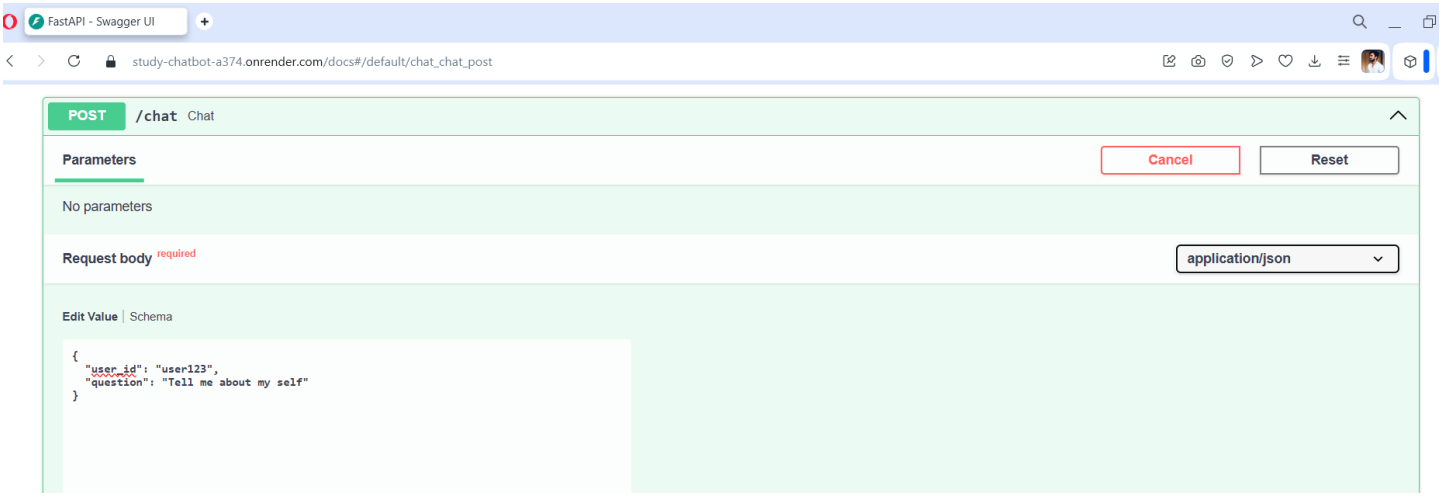
- ⇨ Memory is injected into the LLM prompt as

  ("placeholder", "{history}")

2. Github Repo - https://github.com/vithusan99/study_chatbot

3. Hosted API Link - https://study-chatbot-a374.onrender.com/docs#/default/chat_chat_post

4. Testing API



FastAPI - Swagger UI

study-chatbot-a374.onrender.com/docs#/default/chat_chat_post

Curl

```
curl -X 'POST' \
  'https://study-chatbot-a374.onrender.com/chat' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "user_id": "user123",
  "question": "Tell me about my self"
}'
```

Request URL

```
https://study-chatbot-a374.onrender.com/chat
```

Server response

Code	Details
200	Response body


```
{
  "response": "**Vithushan – A Quick Snapshot**\n\n- **Founder & Leader** – You run an IT company, steering technology strategy, product development, and business growth.  \n- **Musical Entertai
ner** – Passionate about music, you perform live, host events, and blend creative artistry with professional skill.  \n- **Content Creator** – You produce engaging audio-visual or written conten
t that showcases your musical talent and business insights, reaching audiences on social media and other platforms.  \n- **Creative-Tech Hybrid** – Your unique blend of tech acumen and artistic
flair positions you to innovate at the intersection of digital media and live performance.  \n\nFeel free to add any details or adjust the focus—this is just a starting point!"
}
```


Download

Response headers

```
access-control-allow-credentials: true
access-control-allow-origin: *
alt-svc: h3=":443"; ma=86400
cf-cache-status: DYNAMIC
cf-ray: 9d1ddd57d958b2fd-CMB
content-encoding: br
content-length: 449
content-type: application/json
date: Sun,22 Feb 2026 10:40:21 GMT
priority: u=1,i
rndr-id: 585cf1ba-981d-4413
server: cloudflare
server-timing: cfExtPri
vary: Accept-Encoding
x-render-origin-server: uvicorn
```

ORGANIZATION Anith's Org - 2020-06... PROJECT chatbot

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0
- admin
- chatbot
- local
- sample_mflix

users

Cluster0 > chatbot > users

View monitoring | Visualize Your Data

Documents 36 | Aggregations | Schema | Indexes 1 | Validation

Type a query: { field: 'value' } or Generate query

Explain | Reset | Find | Options

ADD DATA | UPDATE | DELETE | EXPORT CODE

25 | 1 – 25 of 38

```
_id: ObjectId('6999a810624e0a83f6f4f725')
user_id : "user123"
role : "assistant"
message : "I'm sorry—I don't have that information. If you'd like, you can tell m…"
timestamp : 2026-02-21T12:41:52.238+00:00
```

```
1  _id: ObjectId('6999a877624e0a83f6f4f726')                    ObjectId
2  user_id : "user123"                                          String
3  role : "user"                                                String
   message : "Im vithushan, im a founder of a it companya and im a musical entertainer,   "   String
             conductiong musical event and content creator,
5  timestamp : 2026-02-21T12:43:35.981+00:00                    Date
```

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0
- admin
- chatbot
- local
- sample_mflix

Cluster0 > chatbot > users

View monitoring | Visualize Your Data

Documents 36 | Aggregations | Schema | Indexes 1 | Validation

Type a query: { field: 'value' } or Generate query

Explain | Reset | Find | Options

ADD DATA | UPDATE | DELETE | EXPORT CODE

25 | 26 – 38 of 38

```
_id: ObjectId('699a68ed491228a393ecbcfb')
user_id : "user999"
role : "user"
message : "tell me about me?"
timestamp : 2026-02-22T02:24:45.080+00:00
```

```
1  _id: ObjectId('699a68ed491228a393ecbcfb')                    ObjectId
2  user_id : "user999"                                          String
3  role : "assistant"                                           String
   message : "Here's a quick snapshot of what I've gathered about you so far:   "   String

             | Detail | Information |
             |--------|-------------|
             | **Name** | Anith |
             | **Country** | Sri Lanka |
             | **Profession** | IT Developer (3 years of experience) |
             | **Current Focus** | • Job search: actively looking for new vacanci
             | **Goal-Setting Style** | • Likes to break big goals into micro-tas
```